6.3.5 Deploying a Lambda function with SAM

To deploy a Lambda function, you need to upload the deployment package to S3. The deployment package is a zip file including your code as well as additional modules. Afterward, you need to create and configure the Lambda function as well as all the dependencies (the IAM role, event rule, and so on). Using SAM in combination with the AWS CLI allows you to accomplish both tasks.

First, you need to create an S3 bucket to store your deployment packages. Use the following command, replacing `$yourname` with your name to avoid name conflicts with other readers:

```
$ aws s3 mb s3://ec2-owner-tag-$yourname
```

Execute the following command to create a deployment package and upload the package to S3. Please note: the command creates a copy of your template at output.yaml, with a reference to the deployment package uploaded to S3. Make sure your working directory is the code directory chapter06 containing the template.yaml and lambda_function.py files:

```
$ aws cloudformation package --template-file template.yaml \
 ➥ --s3-bucket ec2-owner-tag-$yourname --output-template-file output.yaml
```

By typing the following command in your terminal, you are deploying the Lambda function. This results in a CloudFormation stack named `ec2-owner-tag`:

```
$ aws cloudformation deploy --stack-name ec2-owner-tag \
 ➥ --template-file output.yaml --capabilities CAPABILITY_IAM
```